

Assignment: File Handling and Logging in Python

Objective:

The objective of this assignment is to understand file handling in Python and implement proper logging

mechanisms using Python's logging module.

Background:

In real-world applications, logging is essential for debugging, monitoring, and auditing.

This assignment focuses on reading files safely and logging all operations instead of using print statements.

Problem Statement:

Design a Python program that reads a text file and logs file operations, warnings, and errors into a log file.

Functional Requirements:

1. Read a text file line by line.
2. Log the start and completion of file reading.
3. Count total lines and empty lines.
4. Log warnings for empty lines.
5. Log line content with line numbers.
6. Handle file not found and unexpected errors using logging.

Logging Requirements:

- Use Python logging module
- Log file name: file_operations.log

- Include timestamp, log level, and message

Input:

Text file containing multiple lines (may include empty lines).

Output:

A log file containing execution details, warnings, and errors.

Learning Outcomes:

- Understand logging levels
- Handle file exceptions gracefully
- Replace print statements with logging

Bonus Tasks:

- Log file size
- Implement log rotation
- Log execution time